

Article

Green Flashes Observed in Optical and Infrared during an Extreme Electric Storm

Gilbert Green ^{1,2} and Naomi Watanabe ^{1,*}¹ Department of Chemistry and Physics, Florida Gulf Coast University, 10501 FGCU Boulevard South, Fort Myers, FL 33965, USA; gi5596@eagle.fgcu.edu or gggreen@stsci.edu² Space Telescope Science Institution, Baltimore, MD 21218, USA

* Correspondence: nwatanabe@fgcu.edu

Abstract: A strong and fast-moving electrical storm occurred in the Southwest Florida region overnight, from 01:00 UTC on 17 April to 07:00 UTC on 17 April 2023. Video recordings were conducted in the region at Latitude N 26.34° and Longitude W 81.79° for 5 h and 15 min, from 01:45 UTC to 07:00 UTC. The camera captured the flashes transforming from pinkish, violet, blue, and then emerald green in the sky twice: the first colored flash lasted 2.0 s, and the second one lasted 0.5 s. The characteristics of the flashes were analyzed using video images integrated with lightning flash data from the Geostationary Lightning Mapper (GLM). To gain deeper insights into the associated atmospheric conditions, the Advanced Baseline Imager (ABI) was also used to help understand the spectral anomalies. Both events had similarities: the same pattern of changing luminous colors in the optical images and the trajectory of the lightning discharges, showing clusters and horizontal distributions. Event 1 occurred mainly over the ocean and featured more intense storms, heavier rain, and denser, higher cloud-tops compared to Event 2, which occurred inland and involved dissipating storms. Moreover, the group energy detected in Event 1 was an order of magnitude higher than in Event 2. We attribute the wavelength of the recorded colored luminosity to varying atmospheric molecular concentrations, which ultimately contributed to the unique spectral line. In this study, we explore the correlation between colored flashes and specific atmospheric concentrations.

Keywords: spider lightning; cloud lightning; cloud-to-ground lightning; luminosity; optical energy; GLM; thunderstorm; spectrum



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1. Introduction

Lightning activity associated with meteorological and atmospheric conditions is studied continuously with modern technology. Lightning flashes are frequently observable during thunderstorms, and different types of lightning activity are primarily dependent on the change in the number of storms [1]. Refs. [2–4] studied the relationship between cloud-top height and lightning flashes. Ref. [4] specifically examined infrared cloud-top measurements to determine the relationships for regional lightning rates for use in global chemistry models. Differences in lightning generation in thunderstorms over ocean and land were studied by [5–7]. Additionally, the relationship between lightning and thunderstorm updraft growth [8] and between lightning and rainfall rates [5–7] is significant for understanding the connection between lightning activity and associated meteorological conditions.

Incloud (IC) flashes typically initiate at the upper boundary of the negative charge and discharge between the main negative and positive charge regions in cloud [9]. In contrast, positive/negative cloud-to-ground (CG) flashes are typically associated with the main positive/negative charge regions [10,11]. Cloud flashes may manifest as extensive horizontal channels, commonly referred to as “spider lightning”. The term “spider lightning” comes from the branched nature of its discharge path that resembles a spider web

and its gradual development that resembles the crawling motion of a spider [12]. Spider lightning is a visually observable, horizontally stratified discharge that exhibits multiple branches (positive/negative leaders have variable behavior [13,14]) and propagates along the cloud base [15,16]. The propagation path of spider discharges extends tens to hundreds of kilometers and lasts for tens to hundreds of milliseconds [12]. The sustained luminosity observed in spider lightning is attributed to the continuing current along the discharge path. During the dissipating stage of storms, a significant accumulation of space charge occurs within storm clouds, leading to the thickening of the sub-cloud space charge layer. This stratified space charge layer has a significant influence on the discharge path [17,18]. In this stage, the rate of lightning discharges is low [17]. Spider lightning discharges are occasionally observed in conjunction with positive CG lightning flashes [19]. Spider lightning discharges have been documented both during [20] and towards the end of storms [21] in Florida. In cloud regions with similar background radiances, oceanic lightning flashes tend to be larger, brighter, more energetic, and likely to have longer horizontal propagation over the ocean than those over land [22], and these characteristics suggest a physical distinction in the nature of the flashes. In GLM measurements, bright emissions are typically caused by positive CG lightning strokes with high peak currents in long horizontal flashes [23].

Most of the spectra of a return stroke, with temperature of about 30,000 K, of a CG flash are radiated by nitrogen ions (N_2^+) [24], with wavelength ranging from 400 nm to 650 nm [25] and the spectra of a stepped leader, with temperature ranging from 15,000 K to 30,000 K, are radiated by neutral nitrogen (N), N_2^+ , and neutral oxygen (O), for which wavelength ranged from 600 to 1050 nm [26]. The spectral lines of ionized oxygen (O^+) with high excited energies in IC flashes are stronger than those in CG flashes within the same area [27]. Ref. [28] reported that nitrogen molecules (N_2) absorb energy under a high electric field, transitioning from their ground state to a higher energy state known as the transition state, resulting in the emission of light, particularly in the near ultraviolet region of the spectrum. When those N_2 , N_2^+ , N, and O are in an excited state, their emission spectra include many distinct bands within the visible light [29,30]. Nitrogen gas significantly influences the light emitted by electrical discharges in the atmosphere, with most of this light being UV radiation within the 280–405 nm range [31,32]. The light in the pink-violet part of the spectrum is produced when electrons collide with N_2 or nitrogen ions (N^+), and the light in the green part of the spectrum is produced when electrons collide with oxygen atoms (O) [33]. Electric charges in fresh air result in better ionization, leading to the production of more light in the visible region of the spectrum [31]. When electrons collide with atoms and molecules, they cause electrons in them to move to higher orbits, putting the atoms and molecules in an excited state [33]. Excited electrons are unstable and return to their original state, emitting colored light based on the energy difference between the orbits. Higher energy collisions move electrons to farther orbits, emitting shorter wavelength light like pink and violet, while lower energy collisions result in longer wavelength light like red [34].

There are studied colored luminous events: ball lightning, aurora, and green clouds. Ball lightning is a rarely observed phenomenon. It typically appears as a spherical object with a diameter ranging from several tens of centimeters to several meters [35–37], displaying various colors, including red, yellow, white, blue, or occasionally green [35]. Ball lightning predominantly moves horizontally [38] at a speed of a few meters per second [35–37], and its duration of visibility varies from one second to over a minute, with most occurrences lasting less than 10 s [25,38]. This phenomenon tends to occur in the vicinity of an active thunderstorm or in association with cloud-to-ground (CG) lightning [39]. Experiments indicate that ball lightning can occur from both positive and negative lightning flashes [37]. It is reported that the spectra lines of ball lightning include atoms of iron, calcium, and silicon, which are commonly found in soils, as well as atoms of nitrogen and oxygen [25]. They report that during the occurrence of ball lightning, an increase in the presence of O_2 and N_2 , in addition to the three main atoms, was found. While not observing events of ball lightning in these specific video frames, the color spectra observed can suggest

the possibility of this type of event occurring and leading to the result of this colored observation. Aurora is the colored light emission caused by plasma collisions in the ionosphere, where the atmosphere is ionized. Green light is emitted when oxygen atoms are excited by high-energy colliding electrons, while pink or purple light is emitted due to strong collisions with nitrogen molecules and ions, which cause larger orbital differences in electrons [34]. The green clouds (often called derecho) are often associated with upcoming tornadoes or low suspended shelf clouds reflecting ground light [40].

In this study, we investigate the energy and luminosity of lightning discharges, along with the associated meteorological and atmospheric conditions relevant to thunderstorms. We utilize video camera recordings of luminosity captured on 17 April 2023, and analyze lightning activity data reported by the GLM and ABI datasets.

2. Materials and Methods

The observations were conducted in Bonita Spring, Southwest Florida, from 01:00 UTC to 07:00 UTC on 17 April 2023. Figure 1b displays a map of South Florida, the location of Bonita Springs. Figure 1c illustrates a zoomed-in view of Figure 1b, showing the camera's location and its respective field of view. The camera system utilized was an iPhone 13 ProMax (Apple, Naples, FL, USA) in Bonita Spring, Florida. This camera can operate from a rate of 24 to 240 frames per second (fps) with two resolutions, 1920×1080 pixels and 3840×2160 pixels. This camera was positioned at coordinates of 26.3° N and 81.8° W.

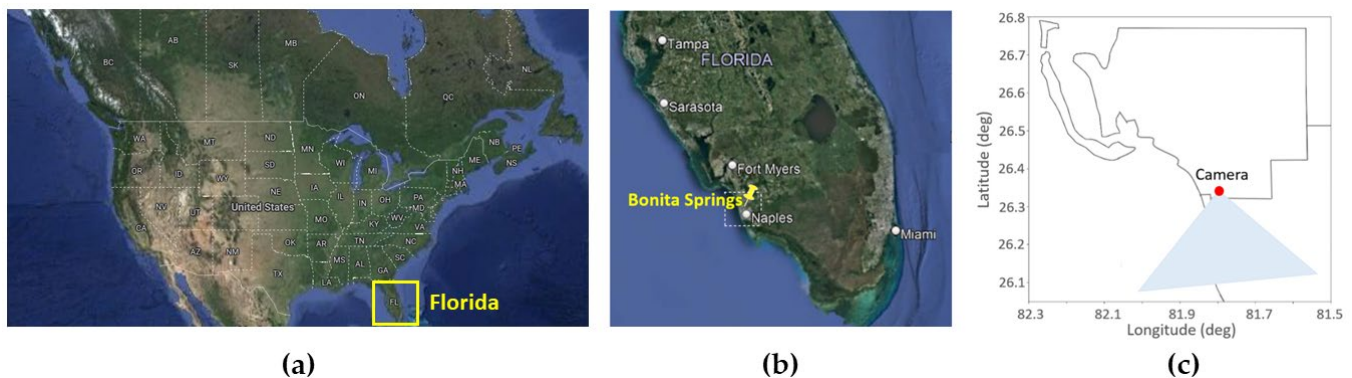


Figure 1. (a) Location of Florida on a map of the United States; (b) a map of Southwest Florida showing the location of Bonita Springs (26.3° N and 81.8° W). (c) Zoomed-in view of the area indicated by the white dot box in (b) showing the location of a camera (26.3° N and 81.8° W) with the red dot and its respective field of view represented by the light blue triangle.

It is equipped with GPS and timestamping to synchronize with the lightning location system data. The GLM data were utilized to analyze and identify locations of specific discharges captured in the recording times and to view the entire propagation path and the total propagation distance and height. The pixel size of the discharge path length and height were measured on the frame, and the actual distance between the camera location and the lightning location reported by the GLM, corresponding to the discharge path that occurred, was determined. These pixel sizes and the actual distance are applied to equations (see Equations (A1) and (A2) in Appendix A) to determine the actual vertical height of the lightning discharge corresponding to the image.

Additionally, the group energy reported by the GLM was used to investigate the intensity of flash luminosity. The GLM is a satellite-based lightning detection system that detects changes in cloud-top illumination and optical energy, specifically within the wavelength range around 777.4 nm, capturing 500 frames per second [8], which corresponds to a 2 ms interval per frame. The GLM detects optical emissions from both cloud lightning and cloud-to-ground lightning flashes [41]. In GLM detection, an event is defined as a pixel on the GLM detecting light brighter than the estimated background brightness during a single 2 ms frame, while groups are clusters of events occurring simultaneously within the

same frame [8,42]. Flashes are larger clusters of groups within a time window of 330 ms or less and a distance within 16.5 km [8], provided they match the Euclidian Distance criterion [42]. Group energy refers to the combined energy and overall footprint of all the events within the group, and the GLM can detect with very low energy, down to 1 femtojoule ($1 \text{ fJ} = 1 \times 10^{-15} \text{ J}$) [43].

In addition to GLM data, Band 13 of the Advanced Baseline Imager (ABI), clean-IR brightness temperature with a longwave of $10.3 \mu\text{m}$, was utilized. ABI is a radiometer capturing IR wavelengths ranging from 0.47 to $13.3 \mu\text{m}$ [44]. The ‘clean’-IR feature is characterized by minimal absorption from ambient water vapor and ozone. This allows for a more precise assessment of convective updraft intensity and glaciation rates with a resolution of 2 km per pixel, covering the continental United States with a grid of 1500 by 2500 pixels [44]. The principal mission of this sensor is viewing clouds in the nighttime; the wavelength that the IR sensor is looking at enables this feature of seeing the complexity of cloud structure when visible light from the sun is not present. The sensor is also used for extreme weather prediction during the daytime, such as hurricanes and strong thunderstorms. The combination of data reported by ABI and GLM helps to understand the characteristics of the cloud-tops in areas where lightning flashes occur. Ref. [45] noted that increasing flash rates correlated with decreasing flash areas, higher cloud-top heights, and colder cloud-top temperatures. However, flash rate and size are more closely associated with the strength of convective storms and precipitation.

We also utilized a chromaticity algorithm to gain insights into the spectra emitted by the flashes observed. The ideal pixels of an image are selected. Red-Green-Blue (RGB) value coordinates of a specific pixel are determined, and the associated color point is determined. The RGB value coordinates are a transformation matrix to convert into x and y chromaticity coordinates; these coordinates and a white point are marked in International Commission on Illumination (CIE) 1931 color space [46]. Utilizing the color space, dominant and complementary wavelengths are determined. The intersection point determines the line to see on the diagram. The line’s interaction with the edges of the diagram determines the dominant and complementary wavelengths. It is important to note that emission lines for specific events do not have one defined wavelength associated with the emission. Therefore, there are typically several spectral lines that are emitted per emission.

We selected data from lightning flashes that occurred within the coordinates ranging from 25.9° N to 26.4° N and from 81° W to 82° W , which sufficiently covered the field of view captured by the cameras from the two observation regions.

3. Results

On 17 April 2023, a regional atmospheric cell, predominantly moving in an easterly direction, stalled and stirred over the Southwest Florida region. A strong and fast-moving storm occurred in the Southwest Florida region overnight, from 01:00 UTC on 17 April (21:00 ET on 16 April) to 07:00 UTC (03:00 ET) on 17 April 2023. During this time, the author observed the same anomalous event three times, where the entire sky turned emerald green, with two of these instances successfully recorded. The video recording duration is 5 h and 15 min, from 01:45 UTC to 07:00 UTC. The video images recorded at 04:13 UTC and 06:34 UTC are analyzed as two events: Event 1 and Event 2. Another event occurring between these times is visually sighted but not documented by video recording or GLM data. Nevertheless, GLM reported fewer lightning events in the same region, and during the specified timeframe, the section may be divided by subheadings. This should provide a concise and precise description of the experimental results and their interpretation, as well as the experimental conclusions that are drawn.

3.1. Event 1 on 17 April 2023 at 04:13 UTC

A lightning event was recorded on 17 April 2023 from 04:13:03.000 UTC to 04:13:05.800 UTC. The camera, located at the coordinates 26.3468° N and 81.7996° W , operated at a frame rate of 24.88 fps, resulting in a frame-to-frame interval of 40.19 ms. A total of

69 frames are recorded, corresponding to a duration of 2.77 s. The time shown in each frame indicates the second within 04:13 UTC (e.g., 4.135 s in the first frame in Figure 2 corresponds to 04:13:04.135 UTC). A very weak luminosity appears on the right side of the image at 2.969 s, and the luminosity starts to brighten at 3.090 s. The luminosity fluctuates between dim and bright repeatedly, with the brighter luminosity occurring from 3.854 s to 4.256 s. Figure 2 illustrates the time from 4.135 s to 4.899 s, corresponding to 04:13:04.135 UTC to 04:13:04.899 UTC, which includes some of the brighter luminosity and different colored luminosities. Among all the frames, only one frame at 4.215 s captures a lightning discharge propagating horizontally, as shown in Figure 2, occurring at 04:13:04.215 UTC. The zoomed-in frame of the visible discharge is presented in Figure 3. As shown in Figure 2, the color of the luminosity changes from pinkish in the frame at 4.577 s to greenish in the next frame at 4.617 s, and then to blueish in the following frames from 4.658 s. Figure 4 displays the GLM lightning events from 04:13:03.500 UTC to 04:13:05.500 UTC. The colored dots represent the lightning locations, with color ranging from red to violet, within a 0.25 s time window. The grayscale illustrates a cloud-top height detected by the longwave IR band (13) of ABI. The first detection during the timeframe was 04:13:03.908 UTC, shown as a red dot in Figure 4. The last discharge in this timeframe occurred at 04:13:05.459 UTC. A cluster of red, orange, and yellow dots is displayed in the lower left of the map, within the coordinates of 25.97° N–26.07° N and 81.93° W–81.81° W during the time interval from 3.5 s to 4.5 s, as shown in Figure 4, and these are assumed to be a single flash [8]. Moreover, the cloud density in the area of the clusters is much lower than in the area above latitude 26.10° N. The average distance of the flash from the camera location is determined to be 42.5 km, using an average distance of the red, orange, and yellow dots reported by GLM. The time of the discharge, shown in Figure 3, corresponds to the orange dot time window, which spans from 04:13:04.000 UTC to 04:13:04.250 UTC within a 250 ms time window, as displayed in Figure 4. Since one frame is about 40 ms, the duration of the visible discharge could occur within 40 ms. The adjacent frames, both before and after the discharge, exhibited significant brightness, indicating the occurrence of a discharge corresponding to the dots. The discharge could be classified as cloud-to-air, cloud-to-cloud, or spider lightning. An interesting aspect of this event, displayed in Figure 4, is the red, orange, yellow, light blue, and green dots occurring from 3.50 s to 4.75 s. These flashes are observed over the ocean, and the first flash on land is represented by the light blue dot occurring at 4.999 s, almost simultaneous with the time window of the blue dots. Moreover, less dense clouds are observed in the region where the red to orange dots are detected, while dense clouds are distributed in the region of the light blue to blue dots.

After the last frame shown in Figure 2, no luminosity is observed in the remaining frames, while blue and violet dots are displayed in Figure 4 during the time window from 4.5 s to 5.0 s. Although these dots are within the camera's field of view, the area occupied by light blue and blue dots is covered by dense cloud, explaining the absence of any visual discharge or brightness in those frames. Figure 5 illustrates the group energy of the lightning discharges reported by GLM from 04:13:03.750 UTC to 04:13:05.500 UTC. Discharges with higher group energy occurred between 04:13:04.000 UTC and 04:13:04.500 UTC, as indicated by the presence of orange and yellow dots in the GLM plot in Figure 4. The group energy higher than 1.00×10^{-13} J was detected as 1.22×10^{-13} J, 1.65×10^{-13} J, and 1.60×10^{-13} J at 4.035 s, 4.037 s, and 4.043 s, respectively, corresponding to the time window of the orange dots. Additionally, a group energy of 1.23×10^{-13} J is detected at 4.271 s, with the highest group energy in the timeframe detected as 2.25×10^{-13} J at 4.436 s, corresponding to the yellow dots. The time of the high group energy of 1.23×10^{-13} J corresponds to the time between the frames of 4.256 s and 4.296 s shown in Figure 2, occurring immediately after the visible discharge and the brighter luminosity in the frames. Figure 6 represents weather maps, which is reported by weather.us [47], corresponding to the area and timeframe of the GLM map shown in Figure 4. The selected times for the weather maps are the closest to and immediately following the timeframe of the GLM map. Figure 6a–c display a storm map at 04:13:48 UTC, which updates every 5 min, a

precipitation map at 05:00 UTC, which updates hourly, and a cloud-top map at 04:15 UTC, which updates approximately every 5 min, respectively. The white circle in Figure 6a–c indicates the area of the cluster of red, orange, and yellow dots during the timeframe from 04:13:03.5 UTC to 04:13:04.5 UTC. By comparing the timeframe and area where the red, orange, and yellow dots occurred in Figure 4 to the area marked with the white circle in Figure 6a–c, it is evident that the intense storm was passing through the area when the dots occurred. During this period, there was an observed increase in precipitation, while the cloud-top height decreased immediately after the occurrence of those dots.

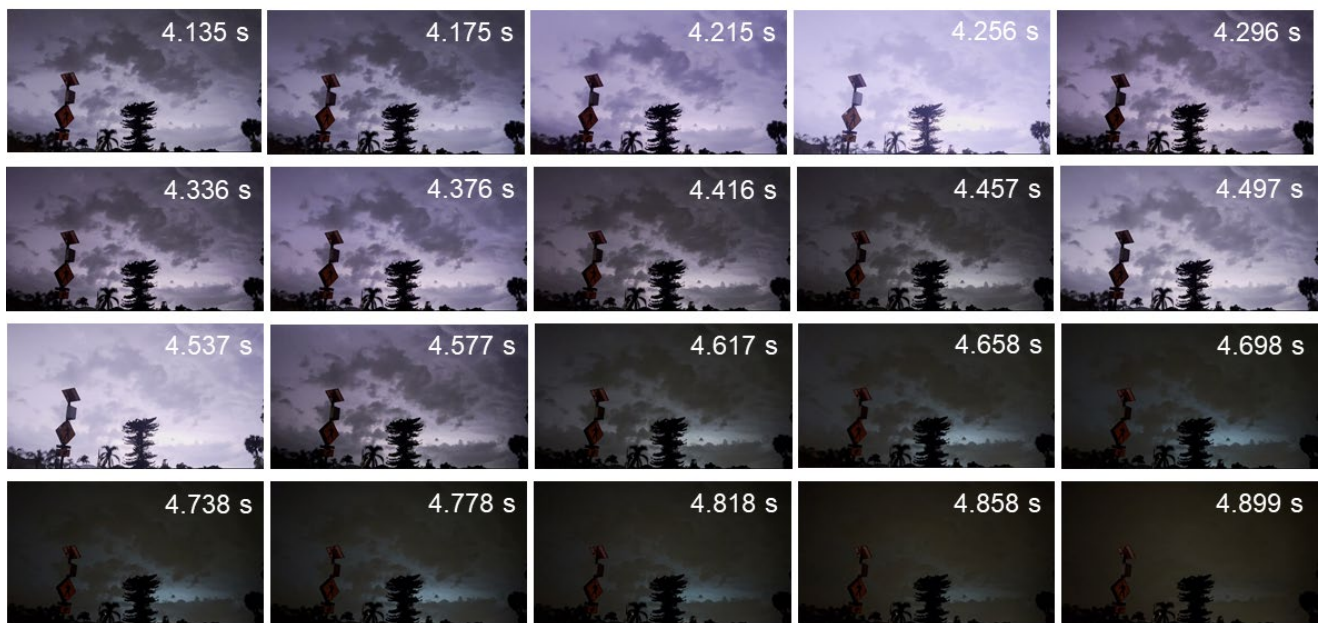


Figure 2. Video-recorded images from 04:13:04.135 UTC to 04:13:04.899 UTC on 17 April. The images are captured by the camera, iPhone 13 ProMax, operated at a frame rate of 24.88 fps, resulting in a frame-to-frame interval of 40.19 ms. The camera's location is the coordinates of 26.3468° N and 81.7996° W.

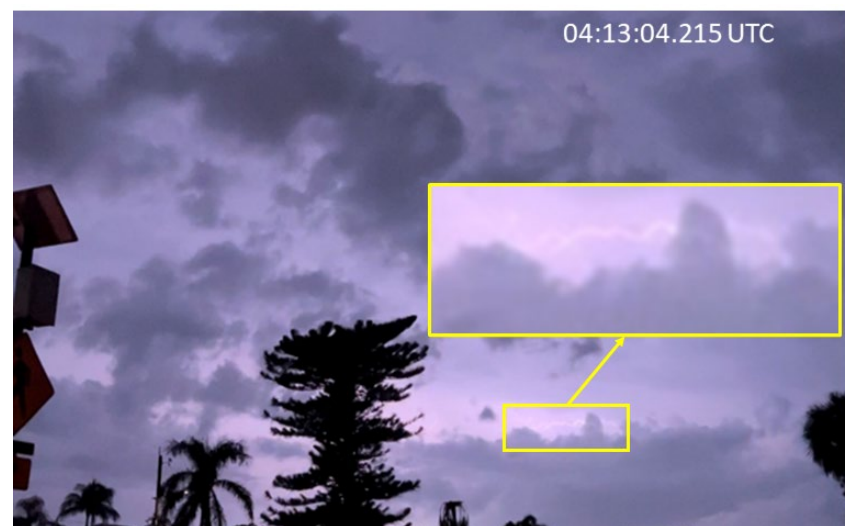


Figure 3. A recorded horizontal discharge at 04:13:04.215 UTC.

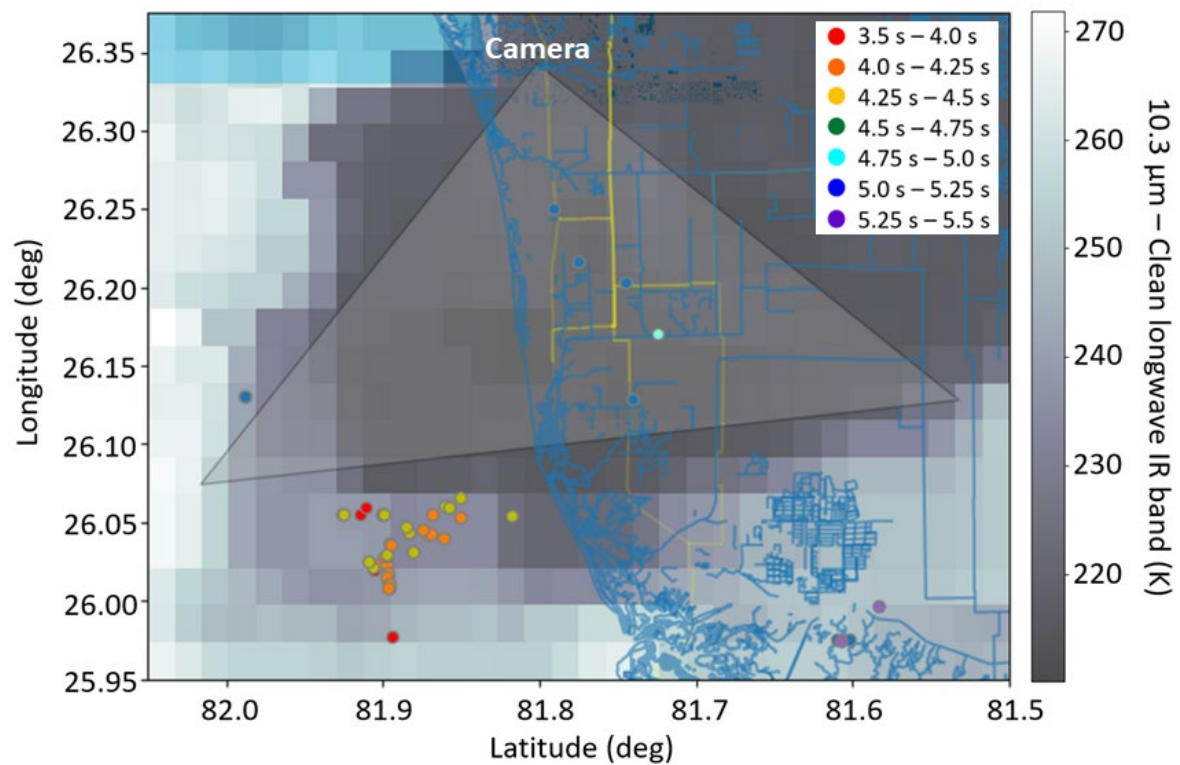


Figure 4. Lightning locations reported by the GLM data on 17 April from 04:13:03.500 UTC to 04:13:05.500 UTC. The map displays a view of the coordinates of 25.95° N–26.35° N and 81.5° W–82.0° W. The colored dots represent the lightning flash locations, with a color dot scale ranging from red to violet within a 0.25 s time window. The total number of lightning detections in each range window is displayed in the box at the upper right corner of the figure. Yellow and blue lines represent transmission lines and river/canal lines, respectively. Clean-IR brightness temperature with longwave 10.3 μm , with units of Kelvin as in absolute temperature.

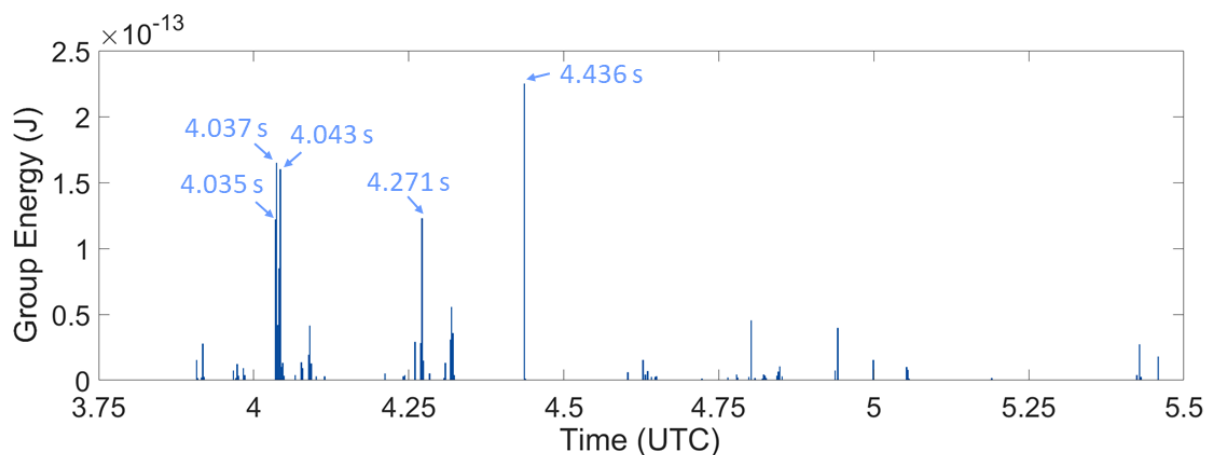


Figure 5. Group energy of lightning discharges reported by GLM from 04:13:03.750 UTC to 04:13:05.500 UTC. The time shown along the horizontal axis indicates the seconds within 04:13 UTC. Given the absence of any detected event from 3.50 s to 3.75 s, the initial time of this plot is set to 3.75 s. Discharges with higher group energy occurred between 04:13:04.0 UTC and 04:13:04.5 UTC, as indicated by the presence of orange and yellow dots in the GLM plot in Figure 4. The highest energy was detected at 04:13:04.4366 s.

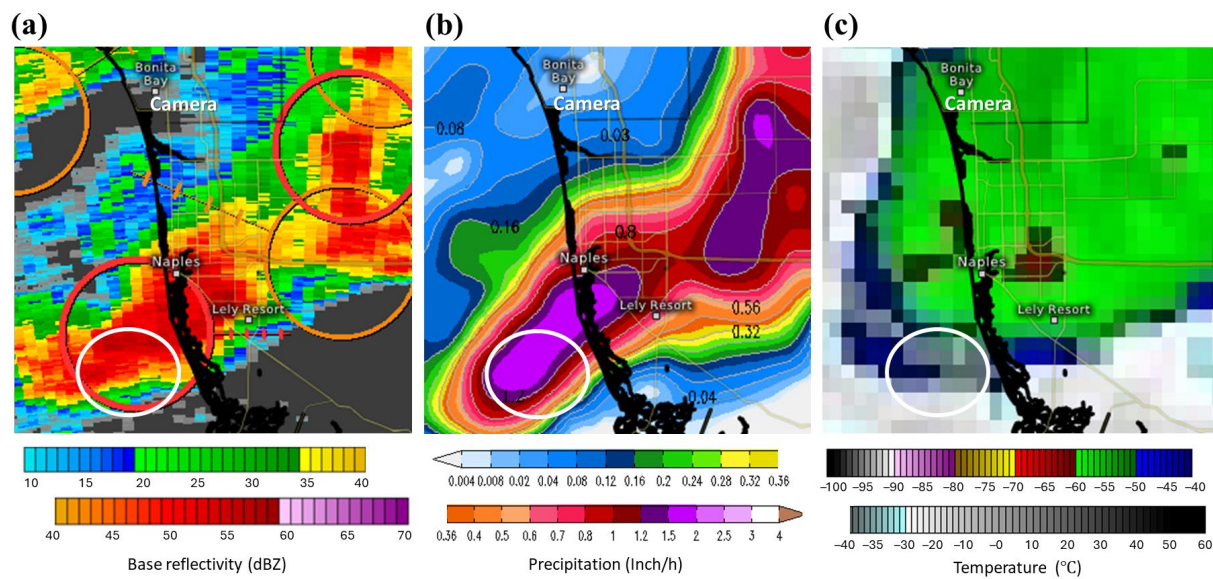


Figure 6. Weather maps reported by [weather.us](#) [47] corresponding to the GLM map in Figure 4. The selected times of the weather maps are the closest to the timeframe of the GLM map. (a) Doppler radar at 04:13:48 UTC (updated every 5 min). The red and orange circles show heavy storm and moderate storm, respectively. (b) Precipitation map at 05:00 UTC (updated every hour). The scale shown below the map represents the total depth of the rainfall from the ground with units of inches (1 inch = 2.54 cm) per hour. (c) Cloud-top map at 04:15 UTC (updated every 5 min). The scale shown below the map represents the temperature of the cloud-top height in degrees Celsius (°C). The white circle in the figure indicates the area of the cluster of red, orange, and yellow dots during the timeframe from 04:13:03.5 UTC to 04:13:04.5 UTC.

3.2. Event 2 on 17 April 2023 at 06:34 UTC

The camera recorded on 17 April 2023 from 06:34:09.6 UTC to 06:34:10.7 UTC. The camera, located at the coordinates 26.3419° N and 81.8013° W, operated at a frame rate of 59.96 fps, resulting in a frame-to-frame interval of 16.67 ms. A total of 61 frames are recorded, corresponding to approximately the duration of 1.02 s. Figure 7 displays frames from 06:34:09.833 UTC to 06:34:10.300 UTC, with the luminosity event lasting for 450 ms, spanning from 9.85 s to 10.30 s. Pinkish luminosity appears at 9.85 s, persists for 117 ms, and then transitions to purple at 9.97 s, where it continues for 100 ms. The final shift from purple to blue occurs at 10.07 s and maintains the same color for a duration of 183 ms. Figure 8 illustrates lightning events reported by GLM on 17 April 2023, with the timeframe ranging from 04:34:09.750 UTC to 04:34:10.750 UTC, covering the observation time of the frames. The map displays coordinates of 26.05° N–26.35° N and 81.5° W–82.05° W, with a color scale ranging from red to violet in a 0.125 s time window. As shown in Figure 8, the discharge is positioned along a straight line, predominantly located beneath the edge of a dense cloud. The first detected event in the map is at 9.850 s.

Figure 9 represents the group energy reported by GLM from 06:34:09.850 UTC to 06:34:10.600 UTC. The higher group energy, exceeding 2.00×10^{-14} J, was detected as 3.06×10^{-14} J, 4.96×10^{-14} J, and 3.86×10^{-14} J at 9.874 s, 9.878 s, and 9.890 s, respectively, corresponding to the frames displaying the pinkish luminosity in Figure 7 and the red and orange dots in Figure 8. The subsequent higher group energy occurs later in the observation time of the luminosity. The highest energy detected in the entire timeframe is 6.92×10^{-14} J at 10.228 s, corresponding to the frame showing the brighter emerald-green luminosity in Figure 7 and the green dot in Figure 8. The following higher group energy was detected as 2.95×10^{-14} J and 3.18×10^{-14} J occurring at 10.256 s and 10.293 s, respectively, corresponding to the frames showing the end of the luminosity in Figure 7 and the emerald-green dots in Figure 8, which are propagating toward the camera. Figure 10 represents weather maps covering the area and the timeframe of the GLM map in Figure 8.

The selected times for the weather maps are the closest to and immediately following the timeframe of the GLM map. Figure 10a–c present a storm map at 06:38:05 UTC, which updates every 5 min, a precipitation map at 07:00 UTC, which updates hourly, and a cloud-top map at 06:35 UTC, which updates every 5 min. The white circle in the figures indicates the area of all dots during the timeframe from 06:34:09.750 UTC to 06:34:10.750 UTC. As displayed in Figure 10a, the heavy storm had already passed the area about 4 min after the event. The previous storm map, updated at 06:33:17 UTC, about a minute before the event, shows a heavy storm in the area, as shown in Figure A3b in Appendix B. If the event occurred right after the heavy storm passed through the area, it supports the statement that the horizontal trajectory of the GLM dots occurred during the storm’s dissipation, indicating spider lightning [17,18]. According to Figure 10b, rain continued during the event time, although the heavy rain had moved away by then. Figure 10c indicates that the cloud-top height decreased during the event.



Figure 7. Video recorded images from Frame 11 (06:34:09.995 UTC) to Frame 53 (04:13:04.899 UTC) on 17 April. The images are captured by the camera, iPhone 13 ProMax, operated at a frame rate of 59.96 fps, resulting in a frame-to-frame interval of 16.67 ms. The camera’s location is the coordinates of 26.3419° N and 81.8013° W.

3.3. Chromatic Analysis of Event 2

For the analysis of chromaticity, two frames—one displaying pinkish/violet luminosity and another with emerald-green/blueish luminosity—are selected as reference frames. A pixel is selected at bright luminosity, most emphasizing these color spectra. Figure 11a represents a reference frame showing the luminosity with pinkish/violet light (at 9.950 s), as shown in Figure 7. The selected pixel coordinates (750, 860) have an RGB value of (112, 74, 97). Figure 11b illustrates a zoomed-in view of this selected pixel. Figure 11c illustrates the CIE 1931 chromaticity diagram. The CIE wavelength of the emitted light from the selected pixel on the image is 440 nm. Figure 11d displays a reference frame with emerald-green/blueish light (at 10.22 s), as shown in Figure 7. The selected pixel coordinates are (800, 880), with an RGB value of (40, 111, 155). Figure 11e illustrates a zoomed-in view of this selected pixel. Figure 11f illustrates the CIE 1931 chromaticity diagram. The CIE wavelength

of the emitted light from the selected pixel on the image is 490 nm. The pinkish/violet pixel, with a wavelength of 420 nm, is very short and close to UV light, influenced by nitrogen (N_2) gas in the atmosphere [26]. On the other hand, the second frame with blueish/green luminous pixels has a wavelength of 490 nm, corresponding to part of the green spectrum produced by electrons colliding with oxygen atoms (O) [33], as indicated in the Introduction section. When the frame shows pinkish/violet luminosity, it indicates that electrons with high energy are colliding with atoms and molecules. These collisions cause the electrons in the atoms and molecules to move to higher energy levels, resulting in the emission of light with shorter wavelengths, which corresponds to the pinkish/violet color [34]. On the other hand, when the frame shows emerald-green/blueish luminosity, it suggests that the energy of the electrons involved in the collisions is lower compared to the energy levels associated with the pinkish/violet luminosity. The concentration of molecules is expected to remain consistent over the short timescale of a lightning discharge (tens of milliseconds). Changes in luminous color during a lightning discharge can be caused by momentary shifts in the discharge's behavior while interacting with atoms and molecules in cloud particles or water droplets. Additionally, abrupt changes in the concentration of these atoms and molecules, influenced by factors such as wind or humidity, can also affect the color of the emitted light.

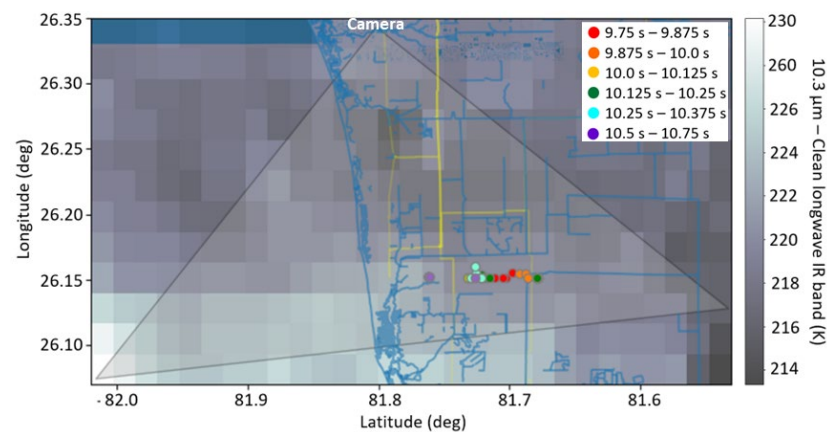


Figure 8. Lightning locations reported by the GLM data on 17 April, from 06:34:09.750 UTC to 06:34:10.750 UTC. The map displays a view of the coordinates of 26.05° N–26.35° N and 81.5° W–82.05° W. The colored dots represent the lightning flash locations, with a color scale ranging from red to violet within a 0.125 s time window. The total number of lightning detections in the time window is displayed in the box at the upper right corner of the figure. Yellow and blue lines represent transmission lines and river/canal lines, respectively. The grayscale indicates Clean-IR brightness temperature with longwave 10.3 μm , with units of Kelvin as in absolute temperature.

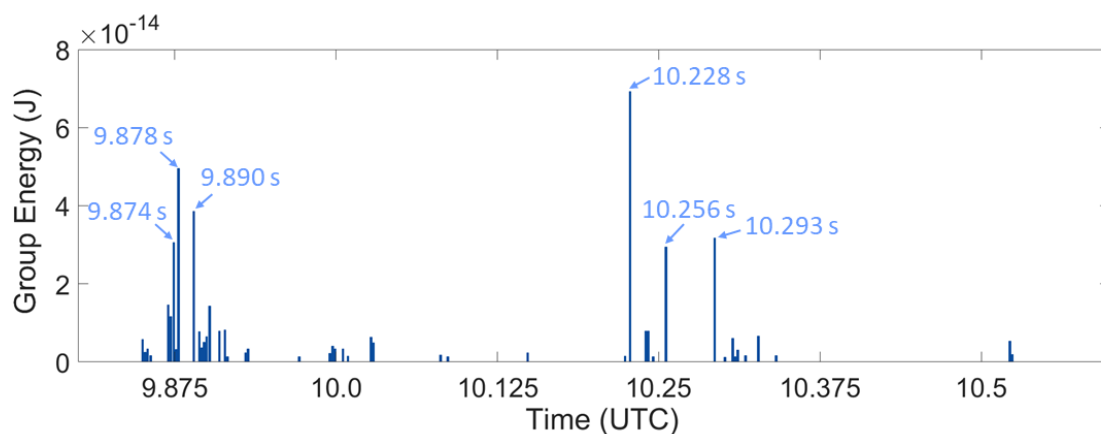


Figure 9. Group energy of lightning discharges reported by GLM from 06:34:09.850 UTC to 06:34:10.600 UTC. The time along the x-axis shows seconds in the timeframe.

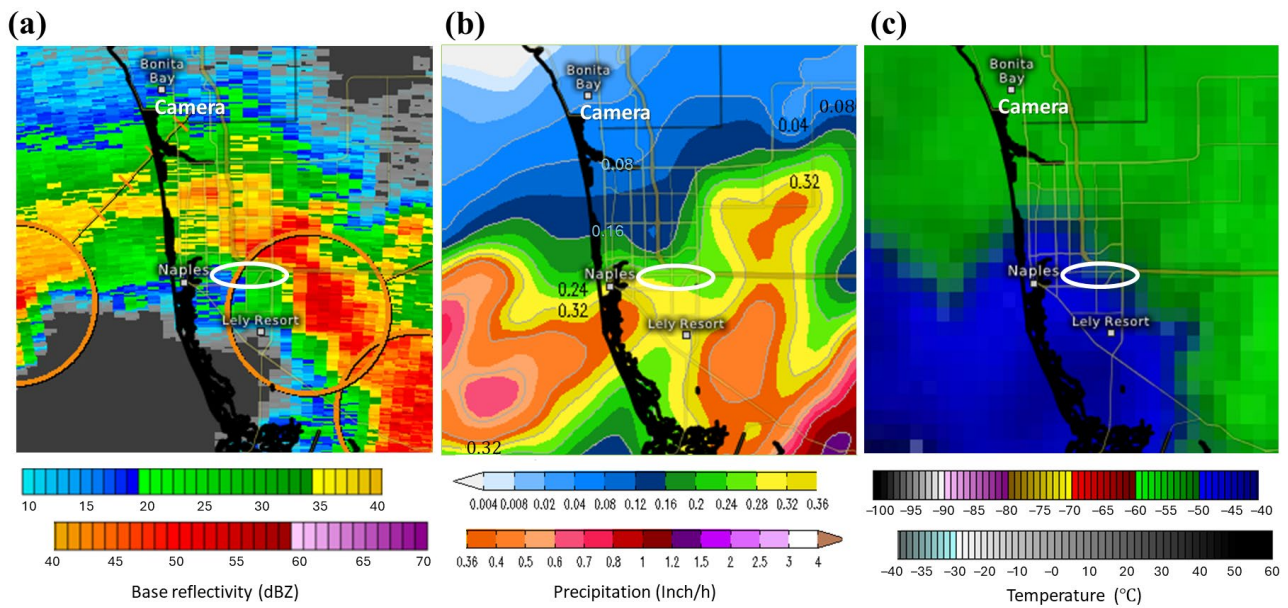


Figure 10. Weather maps reported by weather.us [47] corresponding to the GLM map in Figure 8. The selected times of the weather maps are the closest to the timeframe of the GLM map. (a) Doppler radar at 04:13:48 UTC (updated every 5 min). The orange circle shows a moderate storm. (b) Precipitation map at 05:00 UTC (updated every hour). The scale shown below the map represents the total depth of the rainfall from the ground with units of inches (1 inch = 2.54 cm) per hour. (c) Cloud-top map at 04:15 UTC (updated every 5 min). The scale shown below the map represents the temperature of the cloud-top height in degrees Celsius ($^{\circ}\text{C}$). The white circle in the figure indicates the area of the cluster of red, orange, and yellow dots during the timeframe from 06:34:09.750 UTC to 06:34:10.750 UTC.

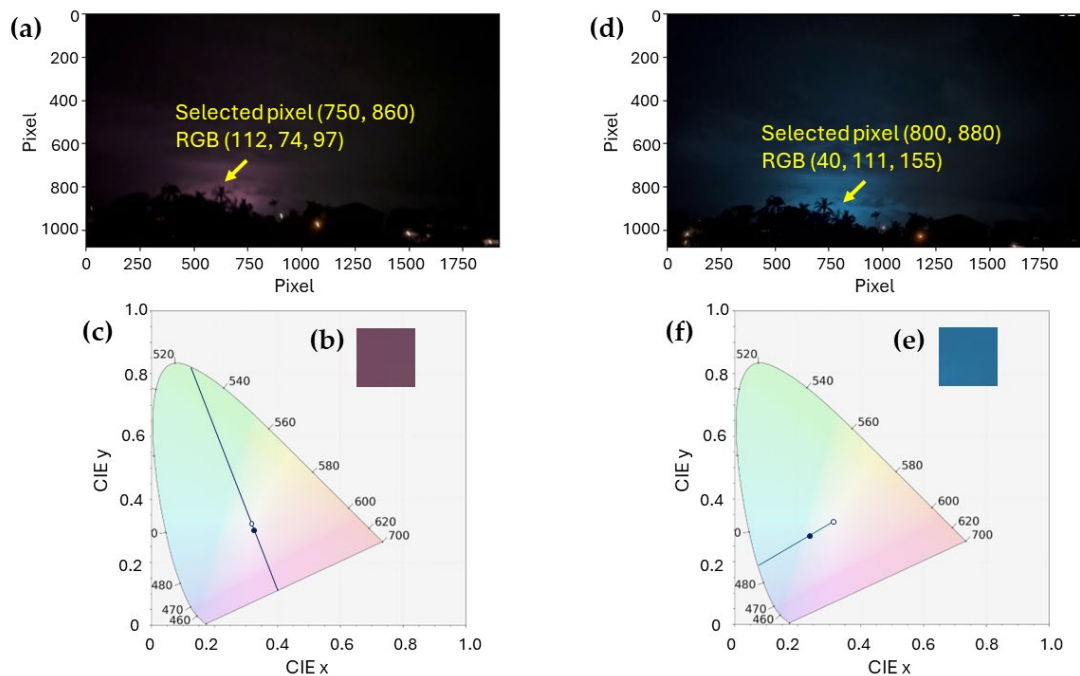


Figure 11. (a) Image of colored luminosity at 9.55 s; (b) zoomed-in pixel image selected in (a); (c) CIE 1931 chromaticity diagram, corresponding to the pixel image in (b); (d) image of colored luminosity at 10.22 s; (e) zoomed-in pixel image selected in (d); (f) CIE 1931 chromaticity diagram, corresponding to the pixel image in (e).

4. Discussion

In Event 1, occurring during 04:13 UTC, a horizontal discharge path is observed in one frame, as shown in Figures 2 and 4, and the corresponding path is detected as red, orange, and yellow dots by GLM, as shown in Figure 4. This horizontal discharge due to the optical image and horizontal trajectory by GLM can support that the image is spider lightning [14]. If so, the spider lightning can propagate below the cloud bottom. The height and total length of the horizontal propagation path recorded in the frame shown in Figure 3 are determined using a specific method, explained in Appendix A. Based on the distance of 42.5 km from the camera to the horizontal discharge, the vertical height of the path was estimated to be 634 m above sea level. The total length of the visible horizontal path, indicated by the yellow box in Figure 3, is determined to be 600 m. Due to image analysis, it was inferred that the horizontal path extended to 1.63 km. The discharge path was captured in only one frame, with the previous frame showing no propagation and the subsequent one exhibiting very bright luminosity, obscuring the path. Several subsequent frames also failed to reveal any discharge path, even though the background had less brightness, indicating that no discharge was captured. Therefore, assuming a total duration of one to two frames ranging from 40 ms to 80 ms and for the horizontal path ranging from 600 m to 1.63 km, the speed of the propagation path is determined to be between 7.5×10^3 – 4.0×10^4 m/s. This speed falls within the range reported in [39], which states that the propagation speed of positive leaders is 1.2 – 4.2×10^4 m/s. The similarity between the estimated speed and the reported range suggests that the observed propagation path may be a positive leader. Based on this estimation, it can be hypothesized that negative charge is located at the cloud base, and the positive leader propagated horizontally beneath the cloud base. Positive leaders are characterized by lower temperature and slower speed compared to negative leaders [17,48].

Figure 12 shows the distribution of the cluster of green and light blue dots occurring very far from the camera at the coordinates of 25.1° N– 26.2° N and 81.5° W– 82.6° W from 4.60 s to 4.94 s, with a duration of 0.330 s. This duration corresponds to the total time of the last eight frames showing the green- and blue-colored luminosity, as represented in Figure 2. As illustrated in Figure 2, the frames display a bluish bright sky during the timeframe of this cluster, with the brightness located on the right side of the image, likely corresponding to the cluster's position in the camera's field of view. If so, the average distance of the cluster of those green and light blue dots is estimated at 153 km from the camera, and the height of the discharge is estimated to be 2.28 km above sea level. When the cluster of green and light blue dots occurred, the conditions of storm, precipitation, and cloud top in the area were similar to those during the timeframe corresponding to the red, orange, and yellow dots (Figure A1a–g in Appendix B; Figure A2a–g in Appendix B).

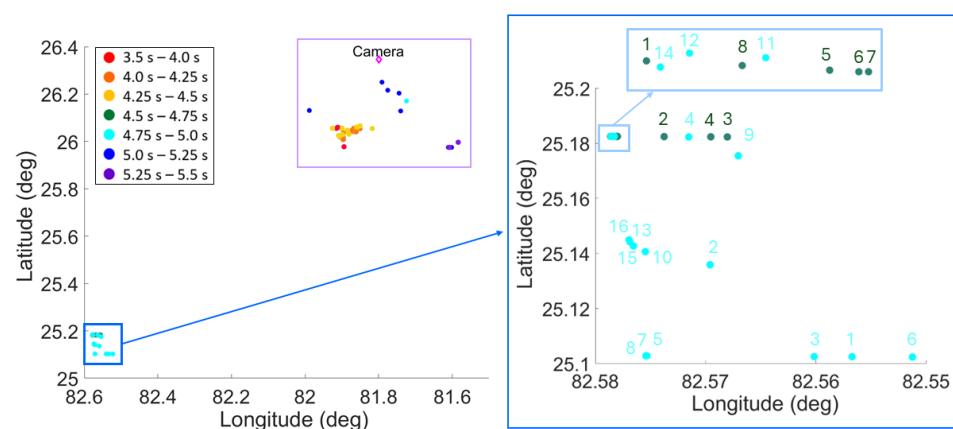


Figure 12. (Left) Extended coordinates of Figure 4 occurring at the coordinates of 25.1° N– 26.2° N and 81.5° W– 82.6° . (Right) Zoomed-in view of the blue box shown in the left figure. The number of each dot represents its chronological order of occurrence.

Oceanic flashes have higher optical power and energy [22], and it is plausible that the oceanic flash occurring in the area transmitted the light event from a considerable distance. Consequently, only green-blue light may have propagated through the clean air above the ocean, or the discharge may have interacted with more oxygen atoms, emitting blue and emerald-green light. This observation may support the presence of blue and emerald-green luminosity captured in the following frames, as shown in Figure 2.

The movement of the cloud towards the camera location supports the hypothesis that the IC flashes occurring within the dense cloud can be observed as colored luminosity. However, the depth of the clouds obscured the flashes, leading to unsuccessful GLM detection. This suggests that the clouds are particularly dense, with cloud particles potentially dispersing light from the lightning flashes. As illustrated in Figure 5, the group energy plot exhibits a pulse-like behavior rather than a gradual increase. This characteristic aligns with the study that there is no gradual rising pattern in the group energy of CG flashes, whereas such a pattern exists in IC flashes [49]. Assuming this pattern, CG may have occurred during the period marked by the green and light blue dots, indicating a potential occurrence of negative CG discharges during that time. Since strokes, particularly the first return strokes, in CGs tend to exhibit strong luminosity at the cloud-top, the GLM has a tendency to detect illumination from other processes in CGs, resulting in higher detection efficiency compared to IC [49]. Additionally, Ref. [49] reported that longer channel lengths and flash durations increase GLM detection efficiency. However, Ref. [50] noted that flashes occurring within storms, which often have short durations, may not always be detected by the GLM. Therefore, it is possible that there were flashes during the observation period that the GLM could not detect.

In Event 2, the trajectory of all the dots from red to violet is almost along a straight line and is located near the edge of the cloud area, as illustrated in Figure 8. These dots are replotted in Figure 13, representing the zoomed-in view of Figure 8, with coordinates of 26.15° N–26.16° N and 81.68° W–81.76° W. The dots within the blue box and the pink box are magnified, respectively. The cluster of orange dots is displayed in the bottom right of the map. The last eight orange dots, ordered from 10 to 17, are closely situated, suggesting that they propagate upward with angle, and the propagation may be branch-like. The location of the yellow, green, and light blue dots indicates that the discharge location is not continuous, which may support the fact that the propagation is not branch-like. Therefore, the event may have exhibited different discharge behaviors during the event time. As shown in Figure 9, higher group energy is detected during the time window corresponding to these different behaviors. This observation supports the hypothesis that discharges with different behaviors exhibit different types of light emission. Assuming a branch-like discharge, there may be more collisions with nitrogen molecules (N_2) or nitrogen ions (N_2^+), resulting in the emission of pinkish or violet light. Conversely, non-branch-like discharges may involve collisions with oxygen atoms (O), emitting blue to emerald-green light. For this event, the discharge path occurred in a time duration of 2.0 s in the same location. The atmospheric condition is considered almost constant within this short timeframe. If so, the change in color of the luminosity could be attributed to the location of the discharge path, with the light dispersed with the interaction between the discharge and unique cloud particles. These interactions may be associated with characteristics of the discharge, such as the temperature of the leader with positive or negative polarity or the composition of molecules at different heights and horizontal distances in the air surrounding the area. During the timeframe of the light blue dots, dots numbered 1 and 2 were located closer to the camera than the other dots. Light blue dot number 2 occurred at 10.228 s, and its energy was the highest during the event, corresponding to the frame of emerald green. The highest energy pulse supports the hypothesis that the high luminous discharge scattered the short-wavelength light of the blue and green onto the molecules in the cloud, resulting in the emerald-green luminosity on the frame. Additionally, this pulse occurs closest to the camera compared to the others, making it easier for the camera to capture the scattered blue and green light.

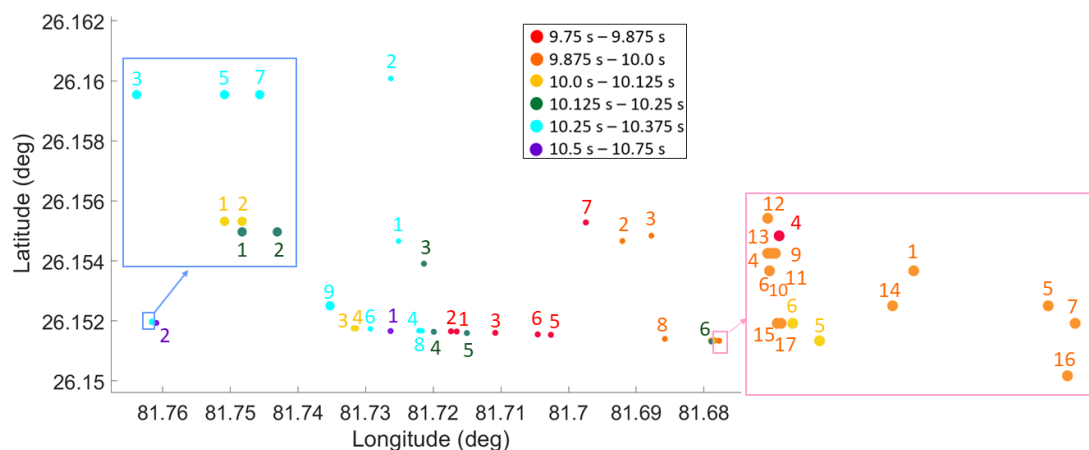


Figure 13. Zoomed-in view of Figure 8, coordinates of 26.15° N–26.162° N and 81.68° W–81.76° W, with a color scale ranging from red to violet within a 0.125 s time window. The dots within the blue box and the pink box are magnified, respectively. The number of each dot represents its chronological order of occurrence.

In comparing the higher and highest energy levels among the six pulses of Event 1 and Event 2, the higher energy of those in Event 1 is about one order greater than those in Event 2. The highest energy in Event 1 is about 3.3 times greater than that in Event 2. During Event 1, there are more intense storms, heavier rain, and denser and higher cloud-tops (See Figures A1 and A2 in Appendix B) compared to Event 2 (See Figure A3 in Appendix B). These differing meteorological conditions are likely to lead to variations in the characteristics of the discharge, even if they are the same type of lightning discharge. Moreover, changes in meteorological conditions can alter atmospheric conditions, thereby affecting the interaction between the discharge and cloud particles in the atmosphere. Consequently, unique wavelengths are transmitted to the camera, resulting in the observation of different colored luminosities.

Both Event 1 and Event 2 produce similar color luminosity patterns; however, Event 1 exhibits less luminosity compared to Event 2. Event 1 involves more intense storms and higher cloud-tops than Event 2. This reduced luminosity in Event 1 may be due to the intense storm causing many molecules and atoms to move away, resulting in lower concentrations in the atmosphere. Consequently, the decreased particle concentration likely reduces the rate of electron collisions, leading to less emission and lower luminosity. In contrast, Event 2, with its weaker storm and lower cloud height, maintains a higher concentration of molecules and atoms, leading to more frequent electron collisions and, thus, more intense-colored luminosity. One assumption is that the storm cleaned up the air, leading to better ionization of the discharge path [31]. Initially, electrons tend to collide with N_2 and N_2^+ , producing emissions of pinkish (light violet) or violet light. Then, after the ionization path is established, electrons start colliding with oxygen atoms, resulting in blue to emerald-green light emission. Another assumption is that the behavior of the discharge changes, such as a change in polarity or direction of the discharge path in a bidirectional leader or recoil leader within the cloud, causes collisions with N_2 or N_2^+ , emitting pinkish (light violet) or violet light, followed by collisions with O, emitting blue to emerald-green light.

Another possibility to consider is explosions occurring at power plants. An anecdotal example that produced similar spectral emissions occurred on 27 December 2018 in Queens, New York City, New York [51]. A Con Edison power plant reportedly experienced an explosion followed by subsequent fires, causing the night sky to light up with blue-green lights for an extended period. To verify this event, we utilized the United States Department of Energy archives of historically reported electrical emergency events and disturbances, Electric Disturbance Events (OE-417) [52]. This event was documented to occur from 9:12 to

9:16 pm ET and caused unexpected transmission loss throughout the area. We consider the possibility of atmospheric electrical interaction with power stations, such as the event at the power plant in New York. During the electric storm on 16–17 April 2023 and the five-hour recording, there were no electrical disturbances reported by OE-417 [52] in the area or other locations in Florida (see Table A1 and Figure A4 in Appendix C). This supports the assumption that the colored luminosity did not result from a power station explosion or fire. Additionally, as indicated in Figures 3 and 8 in Section 3, lightning occurrences were reported during the time of the luminosity by GLM, suggesting a high possibility that the colored luminosity was caused by light emitted from lightning discharges.

The colored luminosity may be caused by the scattering, reflection, and refraction of light by cloud particles. The camera's field of view may have coincidentally been ideally positioned to observe this specific coloration. However, despite multiple events observed by the authors [14] over a couple of years, no other observations of colored luminosity have been reported in the studies. It is also important to note that the time difference between Event 1 and Event 2 (another colored luminosity occurred between Event 1 and Event 2, but it was observed only by eye without recording) was more than two hours. The isolated occurrence of these events suggests that specific lightning flashes interacting with particular atmospheric conditions may have caused these phenomena.

5. Conclusions

On the evening of 17 April 2023, from 01:00 UTC to 07:00 UTC, electrical discharges were observed. Dense cloud systems moved variably, sometimes rapidly and sometimes slowly, through the Southwest Florida area. Winds shifted in accordance with the variable speed of cloud movement. Two events of colored luminosity were documented: Event 1 at 04:13 UTC and Event 2 at 06:34 UTC on the same day, 17 April 2023. In both Event 1 and Event 2, there were similarities in the characteristics observed in the optical images and the GLM maps. Both events recorded the same luminous colors—pinkish (light violet), violet, blue, and emerald green—and the trajectory of the events, as indicated by GLM, showed clusters and horizontal distributions. Event 1 occurred mainly over the ocean, while Event 2 occurred inland. In Event 1, there were more intense storms, heavier rain, and denser and higher cloud-tops compared to Event 2. The group energy detected in Event 1 was an order of magnitude higher than in Event 2.

Both events exhibited a pattern of luminosity changing color within 0.5 to 2 s. Focusing on this short time duration, it is evident that there was no significant change in the atmospheric conditions surrounding the air and the area of the events during this period. If any changes did occur during this timeframe, one possible explanation could be a rapid change in lightning behavior. This may involve interactions between lightning discharge and atmospheric particles, such as molecules, atoms, and ionized particles. Unique concentrations of atmospheric molecular structures may cause the unique monochromatic wavelengths that were observed through visual observation, observation documented by camera systems, and via GLM. Blue shift, unnatural pollutants, and interactions with human power infrastructure systems are also considered. To gain a more comprehensive understanding of the observed color luminosity related to overall lightning activity and thunderstorms, it is necessary to conduct a detailed meteorological analysis, perform spectrum observations to assess atmospheric conditions, and examine cloud flashes using an interferometer. These methods will help determine that atmospheric concentration is the most significant contributor to the visual coloration observed.

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Abbreviations

ET	Eastern Time (3. Results)
GLM	Geostationary Lightning Mapper (Abstract, 1. Introduction, 2. Materials and Method, 3. Results, 4. Discussion, and 5. Conclusions)
ABI	Advanced Baseline Imager (Abstract, 1. Introduction, 2. Materials and Method, and 3. Results)
CG lightning	Cloud-to-ground lightning (1. Introduction and 4. Discussion)
UTC	Coordinated Universal Time (Abstract, 1. Introduction, 2. Materials and Method, 3. Results, 4. Discussion, and 5. Conclusions)
RGB	Red-Green-Blue (2. Materials and Method and 3. Results)
CIE	International Commission on Illumination (2. Materials and Method and 3. Results)
N ₂	Nitrogen molecule (1. Introduction, 3. Results, and 4. Discussion)
N	Neutral nitrogen atom (1. Introduction)
N ₂ ⁺	Nitrogen ion (1. Introduction and 4. Discussion)
O	Neutral oxygen atom (1. Introduction, 3. Results, and 4. Discussion)
O ⁺	Ionized oxygen (1. Introduction)
IC	Incloud lightning flash (1. Introduction and 4. Discussion)
GPS	Global Positioning System (2. Materials and Method)
IR	Infrared (2. Materials and Method)

Appendix A

For the horizontal discharge path appearing in the frame, the pixel size along the horizontal discharge path and its vertical height in the image are measured. The actual distance between the camera and the lightning location reported by the GLM is determined. These pixel sizes and the actual distance are applied to equations (See Equations (A1) and (A2) in Appendix A) to determine the actual vertical height of the lightning discharge corresponding to the image. Additionally, the estimated actual horizontal discharge length is determined using the equations. For the luminosity appearing in the frame, the height of the luminosity is determined by the equations. In this case, where the lightning discharge path is not immediately present, the utility of this formula becomes less reliable. In this case, we choose to rely on GLM observed data and the relative distance from the camera location.

The actual height per pixel, denoted as Y , is determined using Equation (A1):

$$Y = D_y / f \quad (A1)$$

where D represents the distance between the camera and the lightning location, y represents the pixel size of the camera, and f represents the focal length of the camera. The actual height corresponding to the location of the discharge on the image is determined using Equation (A2):

$$H = Yn \quad (A2)$$

where n represents the pixel size of the height of the image.

Appendix B

Figures A1a–g, A2a–g and A3a–f exhibit selected weather maps the closest to and before/after the timeframe of the GLM map.

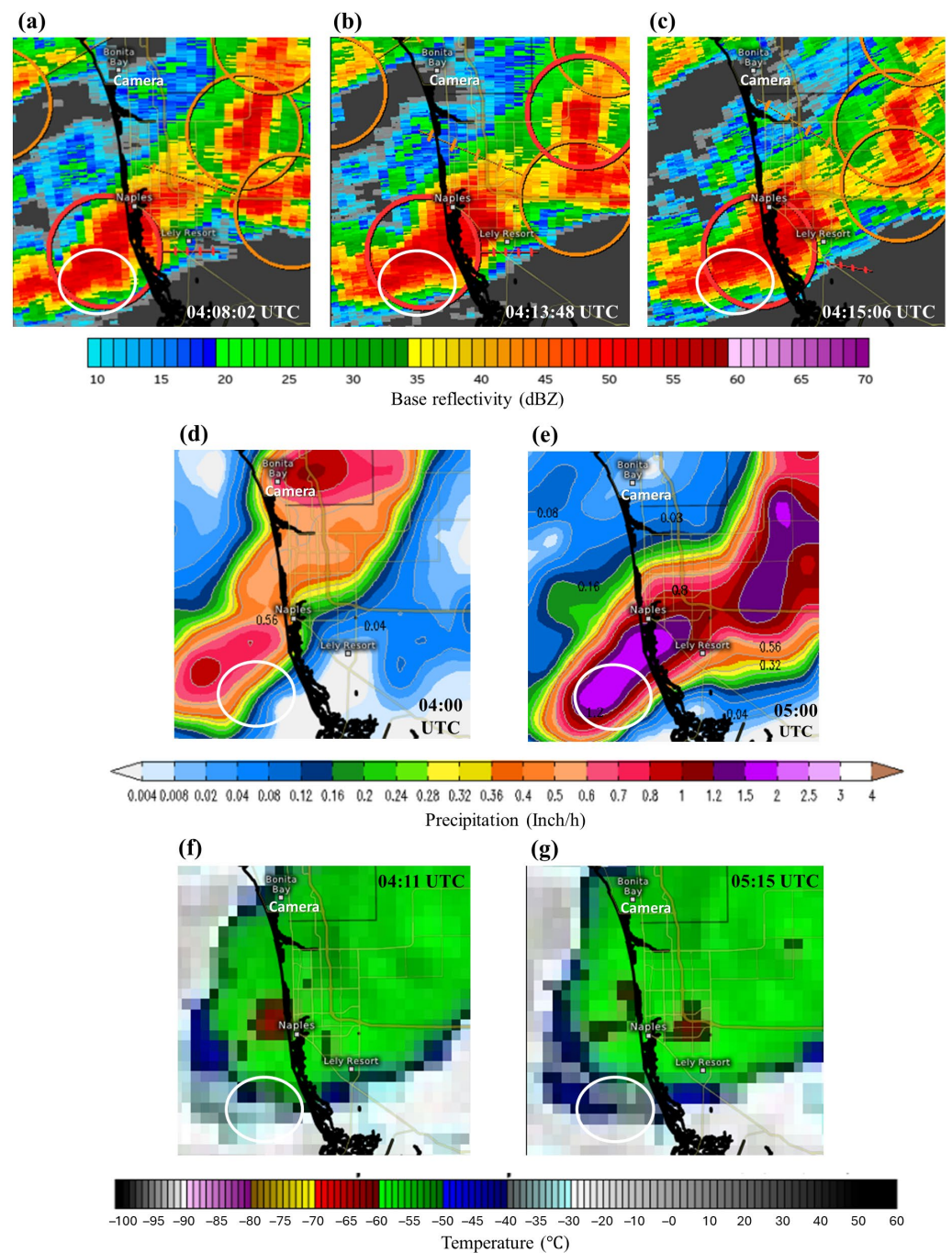


Figure A1. Weather maps reported by weather.us corresponding to the GLM map in Figure 4. The selected times of the weather maps are the closest to and before/after the timeframe of the GLM map. (a) Storm map at 04:08:02 UTC; (b) storm map at 04:13:48 UTC; (c) storm map at 04:15:06 UTC; the storm maps are updated every 5 min. (d) Precipitation map at 04:00 UTC; (e) precipitation map at 05:00 UTC; the precipitation maps are updated every hour. (f) Cloud-top map at 04:11 UTC; (g) cloud-top map at 04:15 UTC; the cloud-top maps are updated approximately every 5 min. The white circle in the figures indicates the area of the cluster of red, orange, and yellow dots during the timeframe from 04:13:03.5 UTC to 04:13:04.5 UTC.

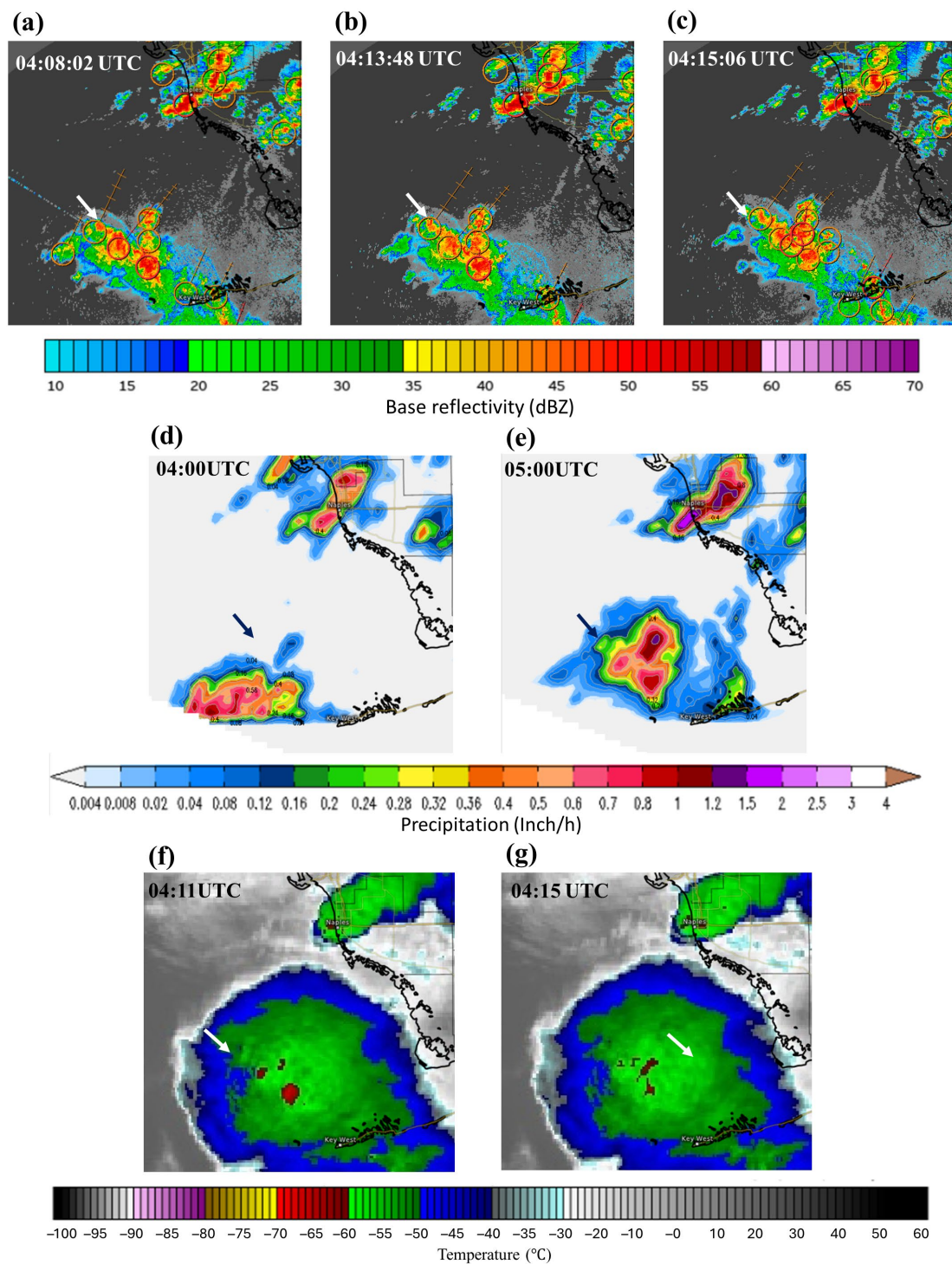


Figure A2. Weather maps corresponding to the GLM map in Figure 4 and the area corresponding to the green and light blue dots in Figure 12. The selected times of the weather maps are the closest to and before/after the timeframe of the GLM map. (a) Storm map at 04:08:02 UTC; (b) storm map at 04:13:48 UTC; (c) storm map at 04:15:06 UTC; the storm maps are updated every 5 min. (d) Precipitation map at 04:00 UTC; (e) precipitation map at 05:00 UTC; the precipitation maps are updated every hour. (f) Cloud-top map at 04:11 UTC; (g) cloud-top map at 04:15 UTC; the cloud-top maps are updated approximately every 5 min. The white arrow in the figures indicates the area of the cluster of green and light blue dots during the timeframe from 04:13:04.5 UTC to 04:13:05.0 UTC. The white arrow in Figures A2a–c and A2f–g, and the black arrow in Figure A2d,e in Appendix B, show the location corresponding to where the light blue dots are detected in Figure 12.

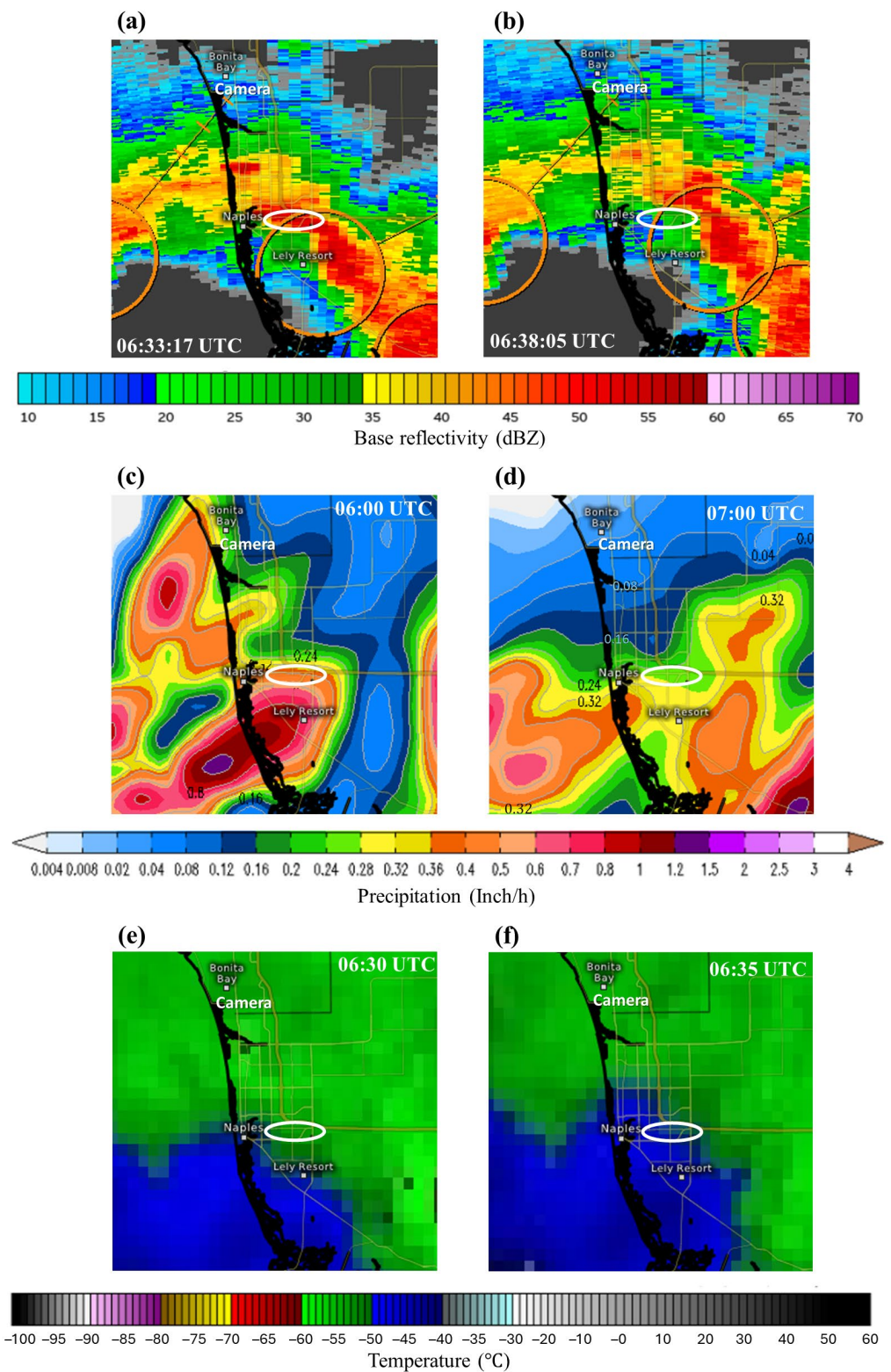


Figure A3. Weather maps corresponding to the GLM map in Figure 8. The selected times of the weather maps are the closest to and before/after the timeframe of the GLM map. (a) Storm map at 06:33:17 UTC; (b) storm map at 06:38:05 UTC; the storm maps are updated approximately every 5 min. (c) Precipitation map at 06:00 UTC; (d) precipitation map at 07:00 UTC; the precipitation maps are updated every hour. (e) Cloud-top map at 06:30 UTC; (f) cloud-top map at 06:35 UTC; the cloud-top maps are updated every 5 min. The white circle in the figures indicates the area of the cluster of all dots during the timeframe from 06:34:09.750 UTC to 06:34:10.750 UTC.

Appendix C

Table A1 indicates the historical reported electrical emergency events and disturbances documented in Electric Disturbance Events (OE-417). Figure A4 shows major power distribution lines in Lee County and Collider County.

Table A1. Historical report of electrical emergency events and disturbances.

Date/Time Event Began	Date/Time of Restoration	Area Affected	Alert Criteria	Event Type
17 April 2023 5:30:00 PM	17 April 2023 2:00:00 PM	Florida: Alachua County;	Damage or destruction of its Facility that results from actual or suspected intentional human action.	Suspicious Activity
17 April 2023 12:17:00 PM	17 April 2023 5:12:00 PM	South Carolina: Aiken County;	Damage or destruction of its Facility that results from actual or suspected intentional human action.	Actual Physical Attack/Vandalism

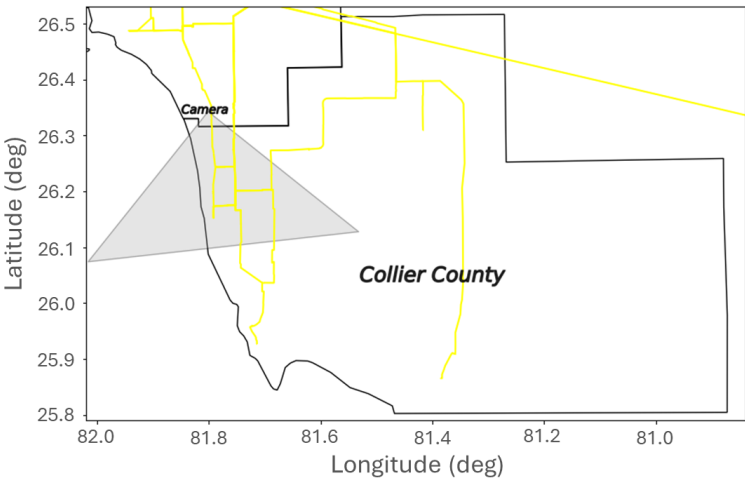


Figure A4. Major power distribution lines in Lee County and Collider County, coordinates of 25.8° N–26.5° N and 81.0° W–82.0° W. The yellow lines illustrate major power distribution lines, and the dark gray triangle represents the camera’s field of view.

References

1. Williams, E.R.; Rothkin, K.; Stevenson, D.; Boccippio, D. Global lightning variations caused by changes in thunderstorm flash rate and by changes in the number of thunderstorms. *J. Appl. Meteorol.* **2000**, *39*, 2223–2230. [\[CrossRef\]](#)

2. Williams, E.R. Large-scale charge separation in thunderclouds. *J. Geophys. Res.* **1985**, *90*, 6013–6025. [\[CrossRef\]](#)

3. Cherna, E.V.; Stansbury, E.J. Sferics rate in relation to thunderstorm dimensions. *J. Geophys. Res.* **1986**, *91*, 8701–8707. [\[CrossRef\]](#)

4. Price, C.; Rind, D. A simple lightning parameterization for calculating global lightning distributions. *J. Geophys. Res.* **1992**, *97*, 9919–9933. [\[CrossRef\]](#)

5. Petersen, W.A.; Rutledge, S.A. On the relationship between cloud-to-ground lightning and convective rainfall. *J. Geophys. Res.* **1998**, *103*, 14025–14040. [\[CrossRef\]](#)

6. Tapia, A.; Smith, J.A.; Dixon, M. Estimation of convective rainfall from lightning observations. *J. Appl. Meteorol. Climatol.* **1998**, *37*, 1497–1509. [\[CrossRef\]](#)

7. Alexander, G.D.; Weinman, J.A.; Karyampudi, V.M.; Olson, W.S.; Lee, A.C.L. The effect of assimilating rain rates derived from satellites and lightning on forecasts of the 1993 superstorm. *Mon. Wea. Rev.* **1999**, *127*, 1433–1457. [\[CrossRef\]](#)

8. Goodman, S.J.; Blakeslee, R.J.; Koshak, W.J.; Mach, D.; Bailey, J.; Buechler, D.; Carey, L.; Schultz, C.; Bateman, M.; McCaul, E., Jr.; et al. The GOES-R geostationary lightning mapper (GLM). *Atmos. Res.* **2013**, *125–126*, 34–49. [\[CrossRef\]](#)

9. Shao, X.M.; Krehbiel, P.R. The spatial and temporal development of intracloud lightning. *J. Geophys. Res. Atmos.* **1986**, *101*, 26641–26668. [\[CrossRef\]](#)

10. Krehbiel, P.R.; Brook, M.; McCrory, R.A. An analysis of the charge structure of lightning discharges to ground. *J. Geophys. Res. Oceans* **1979**, *84*, 2432–2456. [\[CrossRef\]](#)

11. Williams, E.; Mkrtchyan, H.; Mailyan, B.; Karapetyan, G.; Hovakimyan, S. Radar diagnosis of the thundercloud electron accelerator. *J. Geophys. Res. Atmos.* **2002**, *127*, e2021JD035957. [\[CrossRef\]](#)
12. Mazur, V.; Shao, X.M.; Krehbiel, P.R. “Spider” lightning in intracloud and positive cloud-to-ground flashes. *J. Geophys. Res. Atmos.* **1998**, *103*, 19811–19822. [\[CrossRef\]](#)
13. Williams, E.R. Problems in lightning physics—The role of polarity asymmetry. *Plasma Sources Sci. Technol.* **2006**, *15*, S91. [\[CrossRef\]](#)
14. Green, G.; Watanabe, N. Spider Lightning Characterization: Integrating Optical, NLDN, and GLM Detection. *Atmosphere* **2023**, *14*, 1191. [\[CrossRef\]](#)
15. Williams, E.; Zhang, R.; Boccippio, D. Microphysical growth state of ice particles and large-scale electrical structure of clouds. *J. Geophys. Res. Atmos.* **1994**, *99*, 10787–10792. [\[CrossRef\]](#)
16. Boccippio, D.J. The Electrification of Stratiform Anvils. Ph.D. Dissertation, Massachusetts Institute of Technology, Cambridge, MA, USA, 1996.
17. Williams, E.R.; Cooke, C.M.; Wright, K.A. Electrical discharge propagation in and around space charge clouds. *J. Geophys. Res. Atmos.* **1985**, *90*, 6059–6070. [\[CrossRef\]](#)
18. Mazur, V. Physical processes during development of lightning flashes. *Comptes Rendus Phys.* **2002**, *3*, 1393–1409. [\[CrossRef\]](#)
19. Moore, C.B.; Vonnegut, B. The thundercloud. In *Lightning, Physics of Lightning*; Golde, R.H., Ed.; Academic Press: New York, NY, USA, 1997; Volume 1, pp. 51–98.
20. Brantley, R.D.; Tiller, J.A.; Uman, M.A. Lightning properties in Florida thunderstorms from video tape records. *J. Geophys. Res. Atmos.* **1975**, *80*, 3402–3406. [\[CrossRef\]](#)
21. Krider, E.P.; Orville, R.; Noggle, R.; Uman, M. Lightning and the Apollo 17/Saturn V exhaust plume. *J. Spacecr. Rocket.* **1974**, *11*, 72–75. [\[CrossRef\]](#)
22. Peterson, M.; Lay, E. GLM observations of the brightest lightning in the Americas. *J. Geophys. Res. Atmos.* **2020**, *125*, e2020JD033378. [\[CrossRef\]](#)
23. Peterson, M.; Deierling, W.; Liu, C.; Mach, D.; Kalb, C. The properties of optical lightning flashes and the clouds they illuminate. *J. Geophys. Res. Atmos.* **2016**, *122*, 423–442. [\[CrossRef\]](#)
24. Uman, M.A. Determination of lightning temperature. *J. Geophys. Res.* **1969**, *74*, 949–957. [\[CrossRef\]](#)
25. Cen, J.; Yuan, P.; Xue, S. Observation of the optical and spectral characteristics of ball lightning. *Phys. Rev. Lett.* **2014**, *112*, 035001. [\[CrossRef\]](#) [\[PubMed\]](#)
26. Warner, T.A.; Orville, R.E.; Marshall, J.L.; Huggins, K. Spectral (600–1050 nm) time exposures (99.6 μ s) of a lightning stepped leader. *J. Geophys. Res. Atmos.* **2011**, *116*, 116. [\[CrossRef\]](#)
27. Wang, J.; Yuan, P.; Zhang, H.M.; Shen, X.Z. The study on the characteristics and particle densities of lightning discharge plasma. *Guang Pu Xue Yu Guang Pu Fen Xi Guang Pu* **2008**, *28*, 2003–2008.
28. Zhao, Y.; Li, Y.; Li, K.; Han, D.; Qiu, Z.; Zhang, G. Emission Spectrum Analysis of Two Typical Partial Discharge Forms under High Frequency Square Wave Voltages. *IEEE Access* **2020**, *8*, 219946–219954. [\[CrossRef\]](#)
29. Grum, F.; Costa, L.F. Spectral emission of corona discharges. *Appl. Opt.* **1976**, *15*, 76–79. [\[CrossRef\]](#) [\[PubMed\]](#)
30. Janda, M.; Martišovič, V.; Hensel, K.; Machala, Z. Generation of Antimicrobial NO_x by Atmospheric Air Transient Spark Discharge. *Plasma Chem. Plasma Process.* **2016**, *36*, 767–781. [\[CrossRef\]](#)
31. Riba, J.R. Spectrum of corona discharges and electric arcs in air under aeronautical pressure conditions. *Aerospace* **2022**, *9*, 524. [\[CrossRef\]](#)
32. Muhr, M.; Schwarz, R. Experience with optical partial discharge detection. *Mater. Sci.* **2009**, *27*, 1139–1146.
33. Sahin, D. Atomic spectroscopy. In *Modern Spectroscopic Techniques and Applications*; IntechOpen: London, UK, 2019.
34. McLennan, J.C. Bakerian lecture.—The aurora and its spectrum. *Proc. R. Soc. London. Ser. A Contain. Pap. A Math. Phys. Character* **1982**, *120*, 327–357.
35. Shmatov, M.; Stephan, K.D. Advances in ball lightning research. *J. Atmos. Sol. -Terr. Phys.* **2019**, *195*, 105115. [\[CrossRef\]](#)
36. Nikitin, A.I.; Bychkov, V.L.; Nikitina, T.F.; Velichko, A.M.; Abakumov, V.I. Sources and components of ball lightning theory. In *Journal of Physics: Conference Series*; IOP Publishing: Bristol, UK, 2018; Volume 996, p. 012011.
37. Shmatov, M.L. New model and estimation of the danger of ball lightning. *J. Plasma Phys.* **2003**, *69*, 507–527. [\[CrossRef\]](#)
38. Stephan, K.D.; Krajcik, R.; Martin, R.J. Fluorescence caused by ionizing radiation from ball lightning: Observation and quantitative analysis. *J. Atmos. Sol.-Terr. Phys.* **2016**, *148*, 32–38. [\[CrossRef\]](#)
39. Rakov, V.A.; Uman, M.A. *Lightning: Physics and Effects*; Cambridge University Press: Cambridge, UK, 2003.
40. Sneep, M.; Ubachs, W. Direct measurement of the Rayleigh scattering cross section in various gases. *J. Quant. Spectrosc. Radiat. Transf.* **2005**, *92*, 293–310. [\[CrossRef\]](#)
41. Nag, A.; Murphy, M.J.; Schulz, W.; Cummins, K.L. Lightning locating systems: Insights on characteristics and validation techniques. *Earth Space Sci.* **2015**, *2*, 65–93. [\[CrossRef\]](#)
42. Mach, D.M. Geostationary Lightning Mapper clustering algorithm stability. *J. Geophys. Res. Atmos.* **2020**, *125*, e2019JD031900. [\[CrossRef\]](#)
43. Christian, H.J.; Blakeslee, R.J.; Goodman, S.J.; Mach, D.M. *Algorithm Theoretical Basis Document for the Lightning Imaging Sensor*; NASA Earth Science Department Technical Report; NASA: Huntsville, AL, USA, 2000.
44. Schmit, T.J.; Griffith, P.; Gunshor, M.M.; Daniels, J.M.; Goodman, S.J.; Lebar, W.J. A closer look at the ABI on the GOES-R series. *Bull. Am. Meteorol. Soc.* **2017**, *98*, 681–698. [\[CrossRef\]](#)

45. Thiel, K.C.; Calhoun, K.M.; Reinhart, A.E.; MacGorman, D.R. GLM and ABI characteristics of severe and convective storms. *J. Geophys. Res. Atmos.* **2020**, *125*, e2020JD032858. [[CrossRef](#)]
46. Mutschlechner, M.; Schöbel, H. Illuminating Life Sciences: A Biophysical Guide to the Use of Chromatic and White Light Sources in Photobiology. *Photonics* **2024**, *11*, 487. [[CrossRef](#)]
47. Weather.us. Available online: <https://weather.us/> (accessed on 5 June 2024).
48. Lalande, P.; Bondiou-Clergerie, A.; Bacchiega, G.; Gallimberti, I. Observations and modeling of lightning leaders. *Comptes Rendus Phys.* **2002**, *3*, 1375–1392. [[CrossRef](#)]
49. Zhang, D.; Cummins, K.L. Time evolution of satellite-based optical properties in lightning flashes, and its impact on GLM flash detection. *Geophys. Res. Atmos.* **2020**, *125*, e2019JD032024. [[CrossRef](#)]
50. Bruning, E.C.; Rust, W.D.; MacGorman, D.R.; Biggerstaff, M.I.; Schuur, T.J. Formation of charge structures in a supercell. *Mon. Weather. Rev.* **2010**, *138*, 3740–3761. [[CrossRef](#)]
51. Bright Sky at Night: Power Plant Explosion Turns New York City Horizon Blue–Video. 2018. Available online: <https://www.theguardian.com/global/video/2018/dec/28/bright-sky-at-night-power-plant-explosion-turns-new-york-city-horizon-blue-video> (accessed on 5 June 2024).
52. Energy. EO, Electric Disturbance Events (OE-417). Available online: https://www.oe.netl.doe.gov/OE417_annual_summary.aspx (accessed on 5 June 2024).

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